Extraction of Individual Diffusion Coefficients from Two-Layer Gas-Driven Permeation Timelag Measurements

March 24, 2026

Contents

1 Introduction **2**
- 1.1 The Timelag Method . . . 2
- 1.2 Industrial Motivation: Multilayer Barriers . . . 2
- 1.3 Prior Work . . . 2
- 1.4 Contributions and Outline . . . 3

2 Single-Layer Timelag: Classical Derivation **3**
- 2.1 Problem Setup . . . 3
- 2.2 Laplace Transform Solution . . . 4
- 2.3 Downstream Flux and Cumulative Permeation . . . 4
- 2.4 Timelag Extraction via Small-s Expansion . . . 4
- 2.5 Upstream Lead Time . . . 5

3 Two-Layer Composite Timelag Derivation **6**
- 3.1 Problem Setup . . . 6
- 3.2 Laplace Transform in Each Region . . . 7
- 3.3 Applying Boundary and Interface Conditions . . . 7
- 3.4 Downstream Flux and Cumulative Permeation . . . 8
- 3.5 Small-s Expansion and Timelag . . . 9
- 3.6 Direction Dependence of the Downstream Timelag . . . 11
- 3.7 Upstream Lead Time for the Bilayer . . . 12

4 Extraction of Individual Diffusion Coefficients **13**
- 4.1 The Inverse Problem . . . 13
- 4.2 Method 1: Single-Layer Calibration + Bilayer Experiment . . . 14
- 4.3 Method 2: Two Bilayer Samples with Different Thickness Ratios . . . 15
- 4.4 Method 3: Full Transient Curve Fitting . . . 15
- 4.5 Special Case: Known Solubilities . . . 16
- 4.6 Dimensionless Diagnostics and Conditioning . . . 16
  - 4.6.1 The Governing Dimensionless Groups . . . 16
  - 4.6.2 Dimensionless Timelag . . . 17
  - 4.6.3 Regime Analysis . . . 17
  - 4.6.4 The Thin Barrier Coating Case ($\kappa \gg 1$) . . . 18
  - 4.6.5 Practical Diagnostic . . . 18
- 4.7 Summary of Methods . . . 19
- 4.8 Practical Protocol . . . 19

5 Conclusions and Future Work **20**
5.1 Summary of Results 20
5.2 Limitations 21
5.3 Future Directions 21

A Detailed Laplace Transform Calculations **21**
A.1 Simplification of the Coefficient E 22
A.2 Full Small-s Expansion of the Denominator 22
A.3 Verification of the Consistency Check 23
A.4 Reverse-Direction Flux Derivation 23
A.5 Upstream Lead Time: Detailed Expansion 25

1 Introduction

The transport of hydrogen through metallic and polymeric barriers is of central importance in energy systems, chemical processing, and materials engineering. In gas-driven permeation experiments, a membrane of thickness L separates a high-pressure feed chamber from a low-pressure permeate chamber. Hydrogen dissolves into the upstream face of the membrane according to Sieverts' law (metals) or Henry's law (polymers), diffuses down the resulting concentration gradient, and emerges at the downstream face. The cumulative quantity of gas that has permeated, $Q(t)$, is recorded as a function of time.

1.1 The Timelag Method

After an initial transient, $Q(t)$ approaches a linear asymptote whose slope equals the steady-state flux J_{ss} and whose t-intercept defines the *timelag* τ_{lag}. This elegant technique, introduced by Daynes [4] and systematised by Barrer [2], provides a direct route to the diffusion coefficient. For a homogeneous membrane with constant diffusivity D, the classical result is

$$\tau_{\text{lag}} = \frac{L^2}{6D}, \tag{1}$$

requiring only the membrane thickness and the measured intercept.

1.2 Industrial Motivation: Multilayer Barriers

Many practical barriers are not homogeneous single layers. Metallic pipelines may carry oxide scales or deliberately applied coatings; polymer liners in composite pressure vessels can be laminated with barrier films; and cladding strategies in nuclear fuel systems create multilayer diffusion paths. In each case, hydrogen must traverse two or more distinct material layers, each characterised by its own diffusivity D_i and solubility S_i.

When a timelag measurement is performed on such a composite membrane, the resulting τ_{lag} reflects an *effective* combination of the individual layer properties. The fundamental question addressed in this work is:

> *Given experimental permeation data on a bilayer membrane whose layer thicknesses are known but whose individual diffusivities and solubilities are* not*, can the individual transport parameters D_1, D_2, S_1, S_2 be uniquely extracted?*

1.3 Prior Work

The study of diffusion timelags in heterogeneous media has a rich history. Frisch [5] provided early theoretical results on timelags for membranes with position-dependent diffusivity using

variational and moment methods. Ash, Barrer, and Petropoulos [1] derived timelag expressions for laminated slabs, establishing the framework for multilayer analysis. Petropoulos [6] extended the quantitative analysis to glassy polymers. Crank [3] provided comprehensive treatments of diffusion in composite media. Most recently, Tran, Meldon, and Sontag [7] gave a thorough treatment of transient diffusion in bilayer composites, including the important finding that the *downstream* timelag is independent of the direction of permeation when the boundary conditions are ideal (zero concentration at the permeate face).

1.4 Contributions and Outline

In this paper we provide:

1. A self-contained, rigorous derivation of the single-layer timelag via Laplace transforms (section 2), establishing the methodology.
2. A complete derivation of the bilayer composite timelag from the coupled diffusion equations (section 3), including:
   - the full Laplace-domain solution for both layers,
   - the small-s expansion that yields the timelag,
   - a proof that the downstream timelag is direction-independent,
   - derivation of the upstream lead time, which *is* direction-dependent.
3. Systematic extraction methods for the individual transport parameters D_1, D_2, S_1, S_2 from experimentally accessible quantities (section 4), addressing both the case where solubilities are known and the general case where they are not, with analysis of conditioning and practical protocols.

Detailed algebraic manipulations are collected in section A to keep the main exposition readable.

2 Single-Layer Timelag: Classical Derivation

We begin with a rigorous derivation of the classical timelag for a homogeneous membrane, both to establish the Laplace-transform methodology that will be extended to the bilayer case and to derive the upstream lead time that will prove useful for parameter extraction.

2.1 Problem Setup

Consider a homogeneous membrane of thickness L with constant diffusivity D. The concentration $C(x,t)$ satisfies Fick's second law:

$$\frac{\partial C}{\partial t} = D\,\frac{\partial^2 C}{\partial x^2}, \qquad 0 < x < L,\ t > 0. \tag{2}$$

The initial and boundary conditions for a gas-driven permeation experiment are:

$$C(x,0) = 0, \qquad 0 \le x \le L, \tag{3}$$

$$C(0,t) = C_0, \qquad t > 0, \tag{4}$$

$$C(L,t) = 0, \qquad t > 0, \tag{5}$$

where $C_0 = S\sqrt{P}$ for a metal obeying Sieverts' law at the upstream face (or $C_0 = S\,p$ for Henry's law), and the downstream face is maintained at effectively zero concentration by continuous evacuation or sweep gas.

2.2 Laplace Transform Solution

Define the Laplace transform $\bar{C}(x,s) = \mathcal{L}\{C(x,t)\} = \int_0^\infty C(x,t)\, e^{-st}\, \mathrm{d}t$. Transforming eq. (2) with the zero initial condition eq. (3) gives the ODE

$$s\,\bar{C} = D\,\frac{\mathrm{d}^2\bar{C}}{\mathrm{d}x^2}. \tag{6}$$

The general solution is

$$\bar{C}(x,s) = A\cosh(qx) + B\sinh(qx), \qquad q := \sqrt{\frac{s}{D}}. \tag{7}$$

Applying the transformed boundary conditions:

- $\bar{C}(0,s) = C_0/s$ gives $A = C_0/s$.
- $\bar{C}(L,s) = 0$ gives $\dfrac{C_0}{s}\cosh(qL) + B\sinh(qL) = 0$, hence $B = -\dfrac{C_0}{s}\dfrac{\cosh(qL)}{\sinh(qL)}$.

Substituting back:

$$\bar{C}(x,s) = \frac{C_0}{s}\left[\cosh(qx) - \frac{\cosh(qL)}{\sinh(qL)}\sinh(qx)\right] = \frac{C_0}{s}\,\frac{\sinh\big(q(L-x)\big)}{\sinh(qL)}. \tag{8}$$

2.3 Downstream Flux and Cumulative Permeation

The diffusive flux at the downstream face $x = L$ is

$$J(L,t) = -D\left.\frac{\partial C}{\partial x}\right|_{x=L}. \tag{9}$$

In the Laplace domain, differentiating eq. (8):

$$\left.\frac{\partial \bar{C}}{\partial x}\right|_{x=L} = \frac{C_0}{s}\,\frac{-q\cosh\big(q(L-L)\big)}{\sinh(qL)} = \frac{-C_0\,q}{s\,\sinh(qL)}. \tag{10}$$

Therefore,

$$\bar{J}(L,s) = -D\cdot\frac{-C_0\,q}{s\,\sinh(qL)} = \frac{D\,C_0\,q}{s\,\sinh(qL)}. \tag{11}$$

The cumulative permeation is $Q(t) = \int_0^t J(L,t')\,\mathrm{d}t'$, so in the Laplace domain:

$$\bar{Q}(s) = \frac{\bar{J}(L,s)}{s} = \frac{D\,C_0\,q}{s^2\,\sinh(qL)}. \tag{12}$$

2.4 Timelag Extraction via Small-s Expansion

The long-time behaviour of $Q(t)$ corresponds to the small-s behaviour of $\bar{Q}(s)$. We expand systematically. Since $q = \sqrt{s/D}$, we have $qL = L\sqrt{s/D}$ and

$$\sinh(qL) = qL + \frac{(qL)^3}{6} + \frac{(qL)^5}{120} + \cdots = qL\left[1 + \frac{(qL)^2}{6} + \frac{(qL)^4}{120} + \cdots\right]. \tag{13}$$

Substituting $(qL)^2 = sL^2/D$:

$$\sinh(qL) = qL\left[1 + \frac{sL^2}{6D} + \frac{s^2L^4}{120D^2} + \cdots\right]. \tag{14}$$

Now, substituting into eq. (12):

$$\bar{Q}(s) = \frac{D\,C_0\,q}{s^2\,qL\big[1+\frac{sL^2}{6D}+O(s^2)\big]} = \frac{D\,C_0}{s^2\,L}\,\frac{1}{1+\frac{sL^2}{6D}+O(s^2)}. \tag{15}$$

Expanding $\big(1+\epsilon\big)^{-1} \approx 1-\epsilon+\epsilon^2-\cdots$ for small $\epsilon = sL^2/(6D)+O(s^2)$:

$$\bar{Q}(s) = \frac{D\,C_0}{L}\left[\frac{1}{s^2}-\frac{L^2}{6D}\,\frac{1}{s}+O(1)\right]. \tag{16}$$

Inverting term by term using $\mathcal{L}^{-1}\{1/s^2\} = t$ and $\mathcal{L}^{-1}\{1/s\} = 1$:

$$Q(t) \xrightarrow{t\to\infty} \frac{D\,C_0}{L}\left(t-\frac{L^2}{6D}\right) = J_{\rm ss}\,(t-\tau_{\rm lag})\,, \tag{17}$$

where we identify the steady-state flux and timelag:

$$\boxed{J_{\rm ss} = \frac{D\,C_0}{L}, \qquad \tau_{\rm lag} = \frac{L^2}{6D}.} \tag{18}$$

The timelag is the t-intercept of the linear asymptote of $Q(t)$.

2.5 Upstream Lead Time

The *upstream* flux at $x=0$ describes the rate at which gas enters the membrane. From eq. (8):

$$\left.\frac{\partial\bar{C}}{\partial x}\right|_{x=0} = \frac{C_0}{s}\,\frac{-q\cosh(qL)}{\sinh(qL)} = \frac{-C_0\,q\cosh(qL)}{s\,\sinh(qL)}. \tag{19}$$

The upstream flux (into the membrane, taken as positive inward) is

$$\bar{J}_{\rm up}(s) = -D\left.\frac{\partial\bar{C}}{\partial x}\right|_{x=0} = \frac{D\,C_0\,q\,\cosh(qL)}{s\,\sinh(qL)}. \tag{20}$$

The cumulative upstream absorption is

$$\bar{Q}_{\rm up}(s) = \frac{D\,C_0\,q\,\cosh(qL)}{s^2\,\sinh(qL)}. \tag{21}$$

For the small-s expansion, we additionally need

$$\cosh(qL) = 1+\frac{(qL)^2}{2}+\frac{(qL)^4}{24}+\cdots = 1+\frac{sL^2}{2D}+O(s^2). \tag{22}$$

Therefore,

$$\begin{aligned}
\bar{Q}_{\rm up}(s) &= \frac{D\,C_0}{s^2\,L}\,\frac{1+\frac{sL^2}{2D}+O(s^2)}{1+\frac{sL^2}{6D}+O(s^2)} \\
&= \frac{D\,C_0}{s^2\,L}\left[1+\frac{sL^2}{2D}+O(s^2)\right]\left[1-\frac{sL^2}{6D}+O(s^2)\right] \\
&= \frac{D\,C_0}{L}\left[\frac{1}{s^2}+\left(\frac{L^2}{2D}-\frac{L^2}{6D}\right)\frac{1}{s}+O(1)\right] \\
&= \frac{D\,C_0}{L}\left[\frac{1}{s^2}+\frac{L^2}{3D}\,\frac{1}{s}+O(1)\right].
\end{aligned} \tag{23}$$

Inverting:

$$Q_{\rm up}(t) \xrightarrow{t\to\infty} \frac{D\,C_0}{L}\left(t+\frac{L^2}{3D}\right). \tag{24}$$

Remark 2.1. The upstream cumulative absorption increases with a positive offset $+L^2/(3D)$ rather than a negative intercept. This reflects the fact that during the transient, the membrane absorbs gas from the upstream face faster than would occur in steady state. The *upstream lead time* is defined as

$$\boxed{\theta_{\mathrm{up}} = \frac{L^2}{3D} = 2\,\tau_{\mathrm{lag}}.} \tag{25}$$

Remark 2.2. The downstream timelag $\tau_{\mathrm{lag}} = L^2/(6D)$ and upstream lead time $\theta_{\mathrm{up}} = L^2/(3D)$ satisfy $\theta_{\mathrm{up}} = 2\,\tau_{\mathrm{lag}}$. For a single homogeneous layer, both quantities depend on the same single parameter D, so they are not independent. This simple ratio $\theta_{\mathrm{up}}/\tau_{\mathrm{lag}} = 2$ in fact serves as a diagnostic for membrane homogeneity [3, 6]: departures indicate heterogeneity, concentration-dependent diffusivity, or trapping.

3 Two-Layer Composite Timelag Derivation

We now extend the Laplace-transform approach to a bilayer membrane composed of two distinct materials. This section contains the core mathematical contribution: a self-contained derivation of the composite downstream timelag, a proof of its direction independence, and the upstream lead time.

3.1 Problem Setup

Consider a composite membrane consisting of two layers:

- **Layer 1:** $0 < x < \ell_1$, with diffusivity D_1 and solubility S_1.
- **Layer 2:** $\ell_1 < x < \ell_1 + \ell_2$, with diffusivity D_2 and solubility S_2.

The total membrane thickness is $L = \ell_1 + \ell_2$.

In each region, Fick's second law holds:

$$\frac{\partial C_1}{\partial t} = D_1 \frac{\partial^2 C_1}{\partial x^2}, \qquad 0 < x < \ell_1, \tag{26}$$

$$\frac{\partial C_2}{\partial t} = D_2 \frac{\partial^2 C_2}{\partial x^2}, \qquad \ell_1 < x < L. \tag{27}$$

Initial conditions.

$$C_1(x,0) = 0, \qquad C_2(x,0) = 0. \tag{28}$$

Boundary conditions (forward direction).

$$C_1(0,t) = C_0 = S_1\sqrt{P}, \tag{29}$$

$$C_2(L,t) = 0. \tag{30}$$

Interface conditions at $x = \ell_1$.

$$D_1 \left.\frac{\partial C_1}{\partial x}\right|_{x=\ell_1^-} = D_2 \left.\frac{\partial C_2}{\partial x}\right|_{x=\ell_1^+} \qquad \text{(flux continuity)}, \tag{31}$$

$$\frac{C_1(\ell_1^-,t)}{S_1} = \frac{C_2(\ell_1^+,t)}{S_2} \qquad \text{(thermodynamic equilibrium)}. \tag{32}$$

Defining the partition coefficient $\mathrm{K} := S_2/S_1$, the equilibrium condition becomes $C_2(\ell_1^+,t) = \mathrm{K}\,C_1(\ell_1^-,t)$.

3.2 Laplace Transform in Each Region

Transforming eqs. (26) and (27) with zero initial conditions:

$$s\,\bar{C}_1 = D_1\,\frac{\mathrm{d}^2\bar{C}_1}{\mathrm{d}x^2}, \qquad 0 < x < \ell_1, \tag{33}$$

$$s\,\bar{C}_2 = D_2\,\frac{\mathrm{d}^2\bar{C}_2}{\mathrm{d}x^2}, \qquad \ell_1 < x < L. \tag{34}$$

For Layer 1, the general solution is

$$\bar{C}_1(x,s) = A\cosh(q_1 x) + B\sinh(q_1 x), \qquad q_1 := \sqrt{\frac{s}{D_1}}. \tag{35}$$

For Layer 2, it is convenient to introduce the shifted coordinate $y := x - \ell_1$, so that $0 < y < \ell_2$:

$$\bar{C}_2(y,s) = E\cosh(q_2 y) + F\sinh(q_2 y), \qquad q_2 := \sqrt{\frac{s}{D_2}}. \tag{36}$$

3.3 Applying Boundary and Interface Conditions

The four constants A, B, E, F are determined by four conditions.

Condition 1: Upstream boundary. $\bar{C}_1(0,s) = C_0/s$ gives

$$A = \frac{C_0}{s}. \tag{37}$$

Condition 2: Downstream boundary. $\bar{C}_2(\ell_2,s) = 0$ gives

$$E\cosh(q_2\ell_2) + F\sinh(q_2\ell_2) = 0, \tag{38}$$

so that

$$F = -E\,\frac{\cosh(q_2\ell_2)}{\sinh(q_2\ell_2)}. \tag{39}$$

Condition 3: Interface equilibrium. $\mathrm{K}\,\bar{C}_1(\ell_1,s) = \bar{C}_2(0,s)$ gives

$$\mathrm{K}\big[A\cosh(q_1\ell_1) + B\sinh(q_1\ell_1)\big] = E. \tag{40}$$

Condition 4: Interface flux continuity. $D_1\,q_1\big[A\sinh(q_1\ell_1) + B\cosh(q_1\ell_1)\big] = D_2\,q_2\,F$ gives

$$D_1\,q_1\big[A\sinh(q_1\ell_1) + B\cosh(q_1\ell_1)\big] = D_2\,q_2\,F. \tag{41}$$

Solving the system. From eq. (40): $E = \mathrm{K}\big[A\cosh(q_1\ell_1) + B\sinh(q_1\ell_1)\big]$. Substituting into eq. (39):

$$F = -\mathrm{K}\big[A\cosh(q_1\ell_1) + B\sinh(q_1\ell_1)\big]\,\frac{\cosh(q_2\ell_2)}{\sinh(q_2\ell_2)}. \tag{42}$$

Inserting this into eq. (41):

$$D_1\,q_1\big[A\sinh(q_1\ell_1) + B\cosh(q_1\ell_1)\big] = -D_2\,q_2\,\mathrm{K}\big[A\cosh(q_1\ell_1) + B\sinh(q_1\ell_1)\big]\,\frac{\cosh(q_2\ell_2)}{\sinh(q_2\ell_2)}. \tag{43}$$

Define for compactness:

$$R := \frac{D_2\,q_2\,\mathrm{K}\,\cosh(q_2\ell_2)}{\sinh(q_2\ell_2)}. \tag{44}$$

Then eq. (43) becomes, with $A = C_0/s$:

$$D_1\, q_1 \left[\frac{C_0}{s}\sinh(q_1\ell_1) + B\cosh(q_1\ell_1)\right] = -R\left[\frac{C_0}{s}\cosh(q_1\ell_1) + B\sinh(q_1\ell_1)\right]. \quad (45)$$

Solving for B:

$$B = -\frac{C_0}{s}\,\frac{D_1\, q_1\sinh(q_1\ell_1) + R\cosh(q_1\ell_1)}{D_1\, q_1\cosh(q_1\ell_1) + R\sinh(q_1\ell_1)}. \quad (46)$$

3.4 Downstream Flux and Cumulative Permeation

The downstream flux is evaluated at $y = \ell_2$:

$$\bar{J}(L,s) = -D_2\,\frac{\partial\bar{C}_2}{\partial y}\bigg|_{y=\ell_2} = -D_2\, q_2\big[E\sinh(q_2\ell_2) + F\cosh(q_2\ell_2)\big]. \quad (47)$$

Using eq. (39):

$$\begin{aligned} E\sinh(q_2\ell_2) + F\cosh(q_2\ell_2) &= E\sinh(q_2\ell_2) - E\,\frac{\cosh(q_2\ell_2)}{\sinh(q_2\ell_2)}\,\cosh(q_2\ell_2) \\ &= E\,\frac{\sinh^2(q_2\ell_2) - \cosh^2(q_2\ell_2)}{\sinh(q_2\ell_2)} = \frac{-E}{\sinh(q_2\ell_2)}. \end{aligned} \quad (48)$$

Therefore,

$$\bar{J}(L,s) = \frac{D_2\, q_2\, E}{\sinh(q_2\ell_2)}. \quad (49)$$

Now substitute E from eq. (40):

$$E = \mathrm{K}\left[\frac{C_0}{s}\cosh(q_1\ell_1) + B\sinh(q_1\ell_1)\right]. \quad (50)$$

After substituting eq. (46) (the details are given in section A), we obtain:

$$E = \frac{\mathrm{K}\, C_0\, D_1\, q_1}{s\big[D_1\, q_1\cosh(q_1\ell_1) + R\sinh(q_1\ell_1)\big]}. \quad (51)$$

Substituting into eq. (49):

$$\bar{J}(L,s) = \frac{\mathrm{K}\, D_1\, D_2\, q_1\, q_2\, C_0}{s\,\sinh(q_2\ell_2)\big[D_1\, q_1\cosh(q_1\ell_1) + R\sinh(q_1\ell_1)\big]}. \quad (52)$$

Expanding R from eq. (44), the denominator becomes

$$s\big[D_1\, q_1\cosh(q_1\ell_1)\sinh(q_2\ell_2) + \mathrm{K}\, D_2\, q_2\sinh(q_1\ell_1)\cosh(q_2\ell_2)\big], \quad (53)$$

and the flux simplifies to

$$\bar{J}(L,s) = \frac{\mathrm{K}\, D_1\, D_2\, q_1\, q_2\, C_0}{s\big[D_1\, q_1\cosh(q_1\ell_1)\sinh(q_2\ell_2) + \mathrm{K}\, D_2\, q_2\sinh(q_1\ell_1)\cosh(q_2\ell_2)\big]}. \quad (54)$$

The cumulative permeation in the Laplace domain is

$$\bar{Q}(s) = \frac{\bar{J}(L,s)}{s} = \frac{\mathrm{K}\, D_1\, D_2\, q_1\, q_2\, C_0}{s^2\big[D_1\, q_1\cosh(q_1\ell_1)\sinh(q_2\ell_2) + \mathrm{K}\, D_2\, q_2\sinh(q_1\ell_1)\cosh(q_2\ell_2)\big]}. \quad (55)$$

3.5 Small-s Expansion and Timelag

To extract the long-time asymptote of $Q(t)$, we expand eq. (55) for small s. Write $\alpha_i := q_i \ell_i$ with $\alpha_i^2 = s\ell_i^2/D_i$. The relevant expansions to the required order are:

$$\sinh(\alpha_i) = \alpha_i + \frac{\alpha_i^3}{6} + \frac{\alpha_i^5}{120} + \cdots, \tag{56}$$

$$\cosh(\alpha_i) = 1 + \frac{\alpha_i^2}{2} + \frac{\alpha_i^4}{24} + \cdots, \tag{57}$$

$$q_i = \sqrt{s/D_i}, \qquad q_i\,\alpha_i = q_i^2\,\ell_i = \frac{s\,\ell_i}{D_i}. \tag{58}$$

Numerator. Using $q_1 = \sqrt{s/D_1}$ and $q_2 = \sqrt{s/D_2}$:

$$\mathrm{K}\,D_1\,D_2\,q_1\,q_2\,C_0 = \mathrm{K}\,C_0\,\sqrt{D_1\,D_2}\,s. \tag{59}$$

Denominator. Define $\mathcal{D}(s) := D_1\,q_1\cosh(\alpha_1)\sinh(\alpha_2) + \mathrm{K}\,D_2\,q_2\sinh(\alpha_1)\cosh(\alpha_2)$. Expanding each factor:

$$\begin{aligned} D_1\,q_1\cosh(\alpha_1)\sinh(\alpha_2) &= \sqrt{D_1\,s}\left(1 + \frac{s\ell_1^2}{2D_1} + \cdots\right)\left(\alpha_2 + \frac{\alpha_2^3}{6} + \cdots\right) \\ &= \sqrt{D_1\,s}\,\ell_2\sqrt{\frac{s}{D_2}}\left(1 + \frac{s\ell_1^2}{2D_1} + \frac{s\ell_2^2}{6D_2} + \cdots\right) \\ &= \frac{s\,\ell_2\sqrt{D_1}}{\sqrt{D_2}}\left(1 + \frac{s\ell_1^2}{2D_1} + \frac{s\ell_2^2}{6D_2} + \cdots\right). \end{aligned} \tag{60}$$

Similarly,

$$\begin{aligned} \mathrm{K}\,D_2\,q_2\sinh(\alpha_1)\cosh(\alpha_2) &= \mathrm{K}\sqrt{D_2\,s}\left(\alpha_1 + \frac{\alpha_1^3}{6} + \cdots\right)\left(1 + \frac{s\ell_2^2}{2D_2} + \cdots\right) \\ &= \frac{\mathrm{K}\,s\,\ell_1\sqrt{D_2}}{\sqrt{D_1}}\left(1 + \frac{s\ell_1^2}{6D_1} + \frac{s\ell_2^2}{2D_2} + \cdots\right). \end{aligned} \tag{61}$$

Thus,

$$\mathcal{D}(s) = s\left[\frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}} + \frac{\mathrm{K}\,\ell_1\sqrt{D_2}}{\sqrt{D_1}}\right] + s^2\left[\frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}}\left(\frac{\ell_1^2}{2D_1} + \frac{\ell_2^2}{6D_2}\right) + \frac{\mathrm{K}\,\ell_1\sqrt{D_2}}{\sqrt{D_1}}\left(\frac{\ell_1^2}{6D_1} + \frac{\ell_2^2}{2D_2}\right)\right] + O(s^3). \tag{62}$$

Define the leading coefficient:

$$d_0 := \frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}} + \frac{\mathrm{K}\,\ell_1\sqrt{D_2}}{\sqrt{D_1}} = \frac{\ell_2\,D_1 + \mathrm{K}\,\ell_1\,D_2}{\sqrt{D_1\,D_2}}, \tag{63}$$

and the next coefficient:

$$d_1 := \frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}}\left(\frac{\ell_1^2}{2D_1} + \frac{\ell_2^2}{6D_2}\right) + \frac{\mathrm{K}\,\ell_1\sqrt{D_2}}{\sqrt{D_1}}\left(\frac{\ell_1^2}{6D_1} + \frac{\ell_2^2}{2D_2}\right). \tag{64}$$

Assembling the expansion. We have $\bar{Q}(s) = \mathrm{K}\,C_0\,\sqrt{D_1 D_2}\,s/\big[s^2\,(d_0\,s + d_1\,s^2 + \cdots)\big]$, but let us be more careful. Since $\mathcal{D}(s) = s\,d_0 + s^2\,d_1 + O(s^3)$, we have $s^2\,\mathcal{D}(s) = s^3\,d_0 + s^4\,d_1 + \cdots$ and

$$\begin{aligned} \bar{Q}(s) &= \frac{\mathrm{K}\,C_0\,\sqrt{D_1 D_2}\,s}{s^2\big(s\,d_0 + s^2\,d_1 + \cdots\big)} = \frac{\mathrm{K}\,C_0\,\sqrt{D_1 D_2}}{s^2\,d_0\big(1 + s\,d_1/d_0 + \cdots\big)} \\ &= \frac{\mathrm{K}\,C_0\,\sqrt{D_1 D_2}}{d_0}\,\frac{1}{s^2}\left(1 - \frac{d_1}{d_0}\,s + \cdots\right) \\ &= \frac{\mathrm{K}\,C_0\,\sqrt{D_1 D_2}}{d_0}\left[\frac{1}{s^2} - \frac{d_1}{d_0}\,\frac{1}{s} + O(1)\right]. \end{aligned} \tag{65}$$

Inverting term by term:

$$Q(t) \xrightarrow{t\to\infty} \frac{\mathrm{K}\, C_0 \sqrt{D_1 D_2}}{d_0}\left(t - \frac{d_1}{d_0}\right) = J_{\mathrm{ss}}\,(t - \tau_{\mathrm{lag}})\,. \tag{66}$$

Steady-state flux. Using eq. (63):

$$\boxed{J_{\mathrm{ss}} = \frac{\mathrm{K}\, C_0 \sqrt{D_1 D_2}}{d_0} = \frac{\mathrm{K}\, C_0\, D_1\, D_2}{\ell_2\, D_1 + \mathrm{K}\, \ell_1\, D_2}.} \tag{67}$$

Defining the permeability of each layer as $\Phi_i := D_i\, S_i$ and using $\mathrm{K} = S_2/S_1$, $C_0 = S_1\sqrt{P}$, this takes the *series-resistance form*:

$$J_{\mathrm{ss}} = \frac{\sqrt{P}}{\dfrac{\ell_1}{\Phi_1} + \dfrac{\ell_2}{\Phi_2}} = \frac{\sqrt{P}}{\dfrac{\ell_1}{D_1 S_1} + \dfrac{\ell_2}{D_2 S_2}}, \tag{68}$$

which is the driving force $\sqrt{P}$ divided by the total permeation resistance, in direct analogy with resistors in series. Crucially, J_{ss} depends on the *permeabilities* $\Phi_i = D_i S_i$, not on D_i and S_i individually.

Composite timelag. From eq. (65), the timelag is $\tau_{\mathrm{lag}} = d_1/d_0$. Expanding eq. (64) using eq. (63):

$$\begin{aligned} d_1 &= \frac{\ell_1^2 \ell_2}{2\sqrt{D_1 D_2}} + \frac{\ell_2^3 \sqrt{D_1}}{6 D_2^{3/2}} + \frac{\mathrm{K}\, \ell_1^3 \sqrt{D_2}}{6 D_1^{3/2}} + \frac{\mathrm{K}\, \ell_1 \ell_2^2}{2\sqrt{D_1 D_2}} \\ &= \frac{1}{\sqrt{D_1 D_2}}\left[\frac{\ell_1^2 \ell_2}{2} + \frac{\ell_2^3 D_1}{6 D_2} + \frac{\mathrm{K}\, \ell_1^3 D_2}{6 D_1} + \frac{\mathrm{K}\, \ell_1 \ell_2^2}{2}\right]. \end{aligned} \tag{69}$$

Therefore,

$$\tau_{\mathrm{lag}} = \frac{d_1}{d_0} = \frac{\dfrac{\ell_1^2 \ell_2}{2} + \dfrac{\ell_2^3 D_1}{6 D_2} + \dfrac{\mathrm{K} \ell_1^3 D_2}{6 D_1} + \dfrac{\mathrm{K} \ell_1 \ell_2^2}{2}}{\ell_2 D_1 + \mathrm{K} \ell_1 D_2}. \tag{70}$$

This can be rearranged by multiplying numerator and denominator by $1/(D_1 D_2)$:

$$\boxed{\tau_{\mathrm{lag}} = \frac{\dfrac{\ell_1^2 \ell_2}{2 D_1 D_2} + \dfrac{\ell_2^3}{6 D_2^2} + \dfrac{\mathrm{K} \ell_1^3}{6 D_1^2} + \dfrac{\mathrm{K} \ell_1 \ell_2^2}{2 D_1 D_2}}{\dfrac{\ell_2}{D_2} + \dfrac{\mathrm{K} \ell_1}{D_1}}.} \tag{71}$$

Remark 3.1 (Consistency check). Setting $\mathrm{K} = 1$ (i.e., $S_1 = S_2$) and $D_1 = D_2 = D$ in eq. (71):

$$\tau_{\mathrm{lag}} = \frac{\dfrac{\ell_1^2 \ell_2}{2D^2} + \dfrac{\ell_2^3}{6D^2} + \dfrac{\ell_1^3}{6D^2} + \dfrac{\ell_1 \ell_2^2}{2D^2}}{\dfrac{\ell_2}{D} + \dfrac{\ell_1}{D}} = \frac{1}{6D}\,\frac{3\ell_1^2 \ell_2 + \ell_2^3 + \ell_1^3 + 3\ell_1 \ell_2^2}{\ell_1 + \ell_2} = \frac{(\ell_1 + \ell_2)^2}{6D} = \frac{L^2}{6D}, \tag{72}$$

recovering the single-layer result.

Remark 3.2 (Connection to Tran–Meldon–Sontag form). Define the dimensionless parameters $\kappa := \mathrm{K} \ell_1 D_2/(D_1 \ell_2)$ and $\rho := D_1 \ell_2/(D_2 \ell_1)$, and the dimensionless time $\tau_L := \tau_{\mathrm{lag}}\, D_1/\ell_1^2$. Then eq. (71) reduces to the Tran et al. form [7]:

$$\tau_L = \frac{\frac{1}{6} + \frac{\rho^2}{2} + \frac{1}{\kappa}\left(\frac{\rho^2}{6} + \frac{1}{2}\right)}{1 + \frac{1}{\kappa}}, \tag{73}$$

with $\tau_{\mathrm{lag}} = \tau_L\, \ell_1^2/D_1$, confirming agreement.

3.6 Direction Dependence of the Downstream Timelag

Proposition 3.3 (Direction independence of the downstream timelag)**.** *The downstream timelag τ_{lag} for a bilayer membrane with ideal boundary conditions (fixed concentration at the feed face, zero at the permeate face) is independent of the direction of permeation.*

Proof. Consider the **reverse direction**: Layer 2 is now upstream, Layer 1 downstream. The boundary conditions are

$$C_2(L,t) = C_0' = S_2\sqrt{P}, \qquad C_1(0,t) = 0. \tag{74}$$

By symmetry of the derivation in sections 3.2 and 3.3, the reverse-direction Laplace-domain cumulative permeation (measured at the downstream face $x = 0$) has the same structure as eq. (55), but with the roles of the layers exchanged. Specifically, the reverse-direction analogue of $\bar{J}$ at the downstream face ($x = 0$) is obtained by swapping $(\ell_1, D_1, q_1) \leftrightarrow (\ell_2, D_2, q_2)$ and replacing K by 1/K, and $C_0 = S_1\sqrt{P}$ by $C_0' = S_2\sqrt{P}$.

The reverse-direction flux at $x = 0$ in the Laplace domain is (see section A for details):

$$\bar{J}^{\text{rev}}(0,s) = \frac{(1/\mathrm{K})\, D_2\, D_1\, q_2\, q_1\, C_0'}{s\big[D_2\, q_2 \cosh(q_2\ell_2)\sinh(q_1\ell_1) + (1/\mathrm{K})\, D_1\, q_1 \sinh(q_2\ell_2)\cosh(q_1\ell_1)\big]}. \tag{75}$$

Note that $(1/\mathrm{K})\, C_0' = (S_1/S_2)\cdot S_2\sqrt{P} = S_1\sqrt{P} = C_0$. Thus the numerator is identical to the forward case. The denominator bracket becomes

$$D_2\, q_2 \cosh(q_2\ell_2)\sinh(q_1\ell_1) + \frac{D_1\, q_1}{\mathrm{K}}\sinh(q_2\ell_2)\cosh(q_1\ell_1). \tag{76}$$

Comparing with the forward-direction denominator from eq. (53):

$$D_1\, q_1 \cosh(q_1\ell_1)\sinh(q_2\ell_2) + \mathrm{K}\, D_2\, q_2 \sinh(q_1\ell_1)\cosh(q_2\ell_2). \tag{77}$$

These are identical (the two terms are simply swapped in order, and multiplication by K in the forward case corresponds to division by 1/K in the reverse case). More precisely, multiplying the reverse denominator by K:

$$\mathrm{K} D_2\, q_2 \cosh(q_2\ell_2)\sinh(q_1\ell_1) + D_1\, q_1 \sinh(q_2\ell_2)\cosh(q_1\ell_1), \tag{78}$$

which equals the forward denominator. Meanwhile, the reverse cumulative is $\bar{Q}^{\text{rev}}(s) = \bar{J}^{\text{rev}}/s$ with the same overall factor of $(1/\mathrm{K})\cdot C_0' = C_0$ in the numerator.

Since the timelag depends only on the *ratio* d_1/d_0 (where d_0 and d_1 come from expanding the denominator of $\bar{Q}$), and since the denominator function $\mathcal{D}(s)$ is symmetric under the layer-swapping transformation (both directions produce the same $\mathcal{D}$), the timelag is the same in both directions:

$$\tau_{\text{fwd}} = \tau_{\text{rev}}. \tag{79}$$

□

Remark 3.4. The steady-state flux is also direction-independent. In the forward direction, $J_{\text{ss}}^{\text{fwd}} = \mathrm{K}\, C_0\, D_1 D_2/(\ell_2 D_1 + \mathrm{K}\ell_1 D_2)$ with $C_0 = S_1\sqrt{P}$. In the reverse direction, K is replaced by 1/K and C_0 by $C_0' = S_2\sqrt{P}$, giving $J_{\text{ss}}^{\text{rev}} = (1/\mathrm{K})\, C_0'\, D_1 D_2/(\ell_1 D_2 + (1/\mathrm{K})\ell_2 D_1)$. Since $(1/\mathrm{K})\, C_0' = (S_1/S_2)S_2\sqrt{P} = S_1\sqrt{P} = C_0 = \mathrm{K}\, C_0/\mathrm{K}$ and the denominators are identical after regrouping, $J_{\text{ss}}^{\text{fwd}} = J_{\text{ss}}^{\text{rev}}$. In fact, by eq. (68), J_{ss} depends only on $\sqrt{P}$ and the permeabilities $\Phi_i = D_i S_i$, which are material constants independent of direction.

3.7 Upstream Lead Time for the Bilayer

We now derive the upstream cumulative flux at $x = 0$ for the forward direction. The upstream flux into the membrane is

$$J_{\text{up}}(t) = -D_1 \left.\frac{\partial C_1}{\partial x}\right|_{x=0}. \tag{80}$$

From eq. (35) with $A = C_0/s$:

$$\left.\frac{\partial \bar{C}_1}{\partial x}\right|_{x=0} = B\, q_1. \tag{81}$$

Using eq. (46):

$$\bar{J}_{\text{up}}(s) = -D_1\, q_1\, B = \frac{D_1\, q_1\, C_0}{s}\, \frac{D_1\, q_1 \sinh(q_1\ell_1) + R\cosh(q_1\ell_1)}{D_1\, q_1 \cosh(q_1\ell_1) + R\sinh(q_1\ell_1)}. \tag{82}$$

The cumulative upstream absorption is $\bar{Q}_{\text{up}}(s) = \bar{J}_{\text{up}}/s$. For the small-$s$ expansion, we need to expand both numerator and denominator of the fraction in eq. (82).

Numerator of the fraction.

$$\begin{aligned}
&D_1\, q_1 \sinh(q_1\ell_1) + R\cosh(q_1\ell_1) \\
&= D_1\, q_1 \left(\alpha_1 + \frac{\alpha_1^3}{6} + \cdots\right) + \frac{\mathrm{K}D_2\, q_2 \cosh(q_2\ell_2)}{\sinh(q_2\ell_2)} \left(1 + \frac{\alpha_1^2}{2} + \cdots\right) \\
&= \frac{s\ell_1^2}{\ell_1}\left(1 + \frac{s\ell_1^2}{6D_1} + \cdots\right) + \frac{\mathrm{K}D_2\, q_2}{\alpha_2}\left(1 + \frac{\alpha_2^2}{2} + \cdots\right)\left(1 - \frac{\alpha_2^2}{6} + \cdots\right)^{-1} \cdot \alpha_2 \left(1 + \frac{\alpha_1^2}{2} + \cdots\right).
\end{aligned} \tag{83}$$

The detailed expansion (given in full in section A) yields, after careful bookkeeping:

$$\text{Numerator} = \frac{\mathrm{K}D_2}{\ell_2} + s\left[\frac{\ell_1}{} + \frac{\mathrm{K}\ell_2}{3} + \frac{\mathrm{K}D_2\ell_1^2}{2D_1\ell_2}\right] + O(s^2). \tag{84}$$

Denominator of the fraction. This is the same function $\mathcal{D}(s)/s$ that appeared in the downstream problem. From eq. (62):

$$\text{Denominator} = d_0\, s + d_1\, s^2 + O(s^3), \tag{85}$$

but for the upstream cumulative we divide by s^2 overall, giving a different Laurent structure.

After carrying through the full expansion (see section A), the upstream cumulative has the asymptotic form

$$Q_{\text{up}}(t) \xrightarrow{t\to\infty} J_{\text{ss}}\,(t + \theta_{\text{up}})\,, \tag{86}$$

where $\theta_{\text{up}} > 0$ is the upstream lead time. The explicit expression is

$$\boxed{\theta_{\text{up}} = \frac{d_1'}{d_0} - \frac{d_1}{d_0} + 2\,\frac{d_1}{d_0} = \frac{d_1'}{d_0} + \frac{d_1}{d_0} = \tau_{\text{lag}} + \frac{d_1'}{d_0},} \tag{87}$$

where d_1' arises from the numerator expansion of the upstream flux. Explicitly:

$$d_1' = \frac{1}{\sqrt{D_1 D_2}}\left[\frac{\ell_1^2\ell_2}{2} + \frac{\mathrm{K}\ell_2^3 D_1}{3D_2} + \frac{\mathrm{K}\ell_1^2\ell_2}{2}\right]. \tag{88}$$

More explicitly, the upstream lead time is

$$\theta_{\text{up}} = \frac{\dfrac{\ell_1^2\ell_2}{D_1D_2} + \dfrac{\ell_2^3}{3D_2^2} + \dfrac{\mathrm{K}\ell_1^3}{6D_1^2} + \dfrac{\mathrm{K}\ell_1\ell_2^2}{D_1D_2} + \dfrac{\mathrm{K}\ell_1^2\ell_2}{2D_1D_2}}{\dfrac{\ell_2}{D_2} + \dfrac{\mathrm{K}\ell_1}{D_1}}. \tag{89}$$

Proposition 3.5 (Direction dependence of the upstream lead time)**.** *The upstream lead time* θ_{up} *is, in general, direction-dependent:* $\theta_{\text{up}}^{\text{fwd}} \neq \theta_{\text{up}}^{\text{rev}}$.

Proof. The upstream lead time involves the expansion of the upstream flux, which depends on the gradient $\partial C/\partial x$ at the feed face. In the forward direction this is at $x = 0$ (in Layer 1), while in the reverse direction it is at $x = L$ (in Layer 2). The coefficients d_1' involve different combinations of the layer parameters depending on which layer faces the feed. One can verify by direct computation that $\theta_{\text{up}}^{\text{fwd}} = \theta_{\text{up}}^{\text{rev}}$ only when $D_1 = D_2$ and $S_1 = S_2$ (i.e., the homogeneous case). In the general heterogeneous case, the two upstream lead times differ.

As a concrete check, set $\mathrm{K} = 1$, $\ell_1 = \ell_2 = \ell$, and consider $D_1 \neq D_2$. The forward upstream lead time from eq. (89) is

$$\theta_{\text{up}}^{\text{fwd}} = \frac{\frac{\ell^3}{D_1 D_2} + \frac{\ell^3}{3D_2^2} + \frac{\ell^3}{6D_1^2} + \frac{\ell^3}{D_1 D_2} + \frac{\ell^3}{2D_1 D_2}}{\frac{\ell}{D_2} + \frac{\ell}{D_1}}. \tag{90}$$

The reverse upstream lead time is obtained by swapping $D_1 \leftrightarrow D_2$:

$$\theta_{\text{up}}^{\text{rev}} = \frac{\frac{\ell^3}{D_1 D_2} + \frac{\ell^3}{3D_1^2} + \frac{\ell^3}{6D_2^2} + \frac{\ell^3}{D_1 D_2} + \frac{\ell^3}{2D_1 D_2}}{\frac{\ell}{D_1} + \frac{\ell}{D_2}}. \tag{91}$$

These differ unless $D_1 = D_2$, since the terms $1/(3D_2^2) + 1/(6D_1^2) \neq 1/(3D_1^2) + 1/(6D_2^2)$ in general. □

Remark 3.6. The combination of the direction-independent downstream timelag with the direction-dependent upstream lead time provides the foundation for the extraction methods developed in section 4.

4 Extraction of Individual Diffusion Coefficients

Having established the composite timelag, steady-state flux, and upstream lead time expressions, we now address the inverse problem: given experimental measurements on a bilayer membrane, how can the individual transport parameters be determined?

4.1 The Inverse Problem

For a bilayer membrane, four material parameters are unknown a priori:

$$D_1, \quad D_2, \quad S_1, \quad S_2. \tag{92}$$

The layer thicknesses ℓ_1 and ℓ_2 are assumed known from fabrication records or cross-sectional microscopy. The applied upstream pressure P is a controlled experimental quantity.

The key structural observation is that the observables separate into two groups:

1. **Time-domain quantities** (τ_{lag}, θ_{up}) depend on D_1, D_2, and the partition coefficient $\mathrm{K} = S_2/S_1$, but *not* on S_1 or S_2 individually (eqs. (71) and (89)).

2. **Flux quantities** (J_{ss}) depend on the permeabilities $\Phi_i = D_i S_i$ (eq. (68)), coupling diffusivity and solubility.

Remark 4.1 (Direction independence). By theorem 3.3, the downstream timelag τ_{lag} is the same in both directions. The steady-state flux J_{ss} is likewise direction-independent (eq. (68)). Consequently, reversing the sample orientation yields no new information from τ_{lag} or J_{ss}. The upstream lead time θ_{up} *is* direction-dependent (theorem 3.5), but still depends only on D_1, D_2, K.

Counting equations and unknowns. From a single permeation experiment, we can extract at most three independent quantities: τ_{lag}, J_{ss}, and (if the upstream transient is measured) θ_{up}. These provide:

- $\tau_{\text{lag}} = f(D_1, D_2, \text{K})$ (1 equation in 3 unknowns)
- $\theta_{\text{up}} = g(D_1, D_2, \text{K})$ (1 equation in same 3 unknowns)
- $J_{\text{ss}} = h(\Phi_1, \Phi_2) = h(D_1 S_1, D_2 S_2)$ (1 equation in 2 additional unknowns)

With four unknowns (D_1, D_2, S_1, S_2) and at most three equations from one experiment, the system is *under-determined*. Additional information is required, and we describe several strategies below.

4.2 Method 1: Single-Layer Calibration + Bilayer Experiment

The most direct approach is to independently characterise one of the two materials in a separate single-layer permeation experiment.

Step 1: Single-layer experiment on material i. A single-layer sample of material i (thickness ℓ_i^*, where the asterisk denotes the calibration sample) yields:

$$D_i = \frac{(\ell_i^*)^2}{6\,\tau_i^*}, \tag{93}$$

$$\Phi_i = \frac{J_{\text{ss}}^*\,\ell_i^*}{\sqrt{P}}, \qquad \text{whence} \qquad S_i = \frac{\Phi_i}{D_i}. \tag{94}$$

This determines D_i and S_i for one material.

Step 2: Bilayer experiment. With D_i and S_i known, the bilayer timelag and flux provide two equations in the two remaining unknowns D_j and S_j ($j \neq i$).

From eq. (68):

$$\Phi_j = D_j S_j = \frac{\ell_2}{\dfrac{\sqrt{P}}{J_{\text{ss}}} - \dfrac{\ell_1}{\Phi_1}}. \tag{95}$$

(Here we assumed $i = 1$; the formula for $i = 2$ is analogous.)

This determines Φ_j. To separate D_j and S_j, we use the timelag. With $\text{K} = S_2/S_1$ and $\Phi_j = D_j S_j$, the composite timelag eq. (71) becomes a single equation in D_j (since $\text{K} = \Phi_j/(D_j S_i) \cdot (S_i/S_j)$ reduces to known quantities once D_j is fixed).

More explicitly, define $\text{K} = S_j S_i^{-1} = \Phi_j/(D_j S_i)$ (for $j = 2$, $i = 1$). Substituting into eq. (71) gives a single nonlinear equation in the one unknown D_j, which can be solved numerically. Once D_j is found, $S_j = \Phi_j/D_j$.

Proposition 4.2 (Solvability of Method 1)**.** *Given the single-layer calibration of material i and the bilayer measurements τ_{lag} and J_{ss}, the remaining parameters D_j and S_j are generically determined uniquely (subject to the physical constraint $D_j > 0$, $S_j > 0$).*

Proof sketch. The flux equation fixes $\Phi_j > 0$. With Φ_j known, $\text{K} = \Phi_j/(D_j S_i)$ is a decreasing function of D_j. Substituting into the timelag formula yields a continuous function $\tau_{\text{lag}}(D_j)$ on $(0, \infty)$. Since $\tau_{\text{lag}} \to \infty$ as $D_j \to 0$ and $\tau_{\text{lag}} \to \ell_i^2/(6D_i)$ as $D_j \to \infty$ (the fast-layer limit), the intermediate value theorem guarantees at least one solution. Monotonicity of $\tau_{\text{lag}}(D_j)$ on the feasible domain (which can be verified by examining the derivative) ensures uniqueness. □

Remark 4.3. Method 1 requires a separate single-layer sample of one material. This is often available in practice: for hydrogen permeation barriers, the *substrate* material (e.g., a steel) is typically well-characterised, and only the *coating* or *liner* is novel.

4.3 Method 2: Two Bilayer Samples with Different Thickness Ratios

When single-layer calibration is not feasible (e.g., the individual materials cannot be fabricated as free-standing membranes), one can fabricate two bilayer samples of the *same material pair* but with different thickness ratios.

Setup. Let sample k ($k = 1, 2$) have thicknesses $\ell_1^{(k)}$ and $\ell_2^{(k)}$, with $\ell_1^{(1)}/\ell_2^{(1)} \neq \ell_1^{(2)}/\ell_2^{(2)}$. From each sample, measure $\tau_{\text{lag}}^{(k)}$ and $J_{\text{ss}}^{(k)}$. This gives four equations:

$$\tau_{\text{lag}}^{(k)} = f\Big(D_1, D_2, \mathrm{K};\, \ell_1^{(k)}, \ell_2^{(k)}\Big), \quad k = 1, 2, \tag{96}$$

$$J_{\text{ss}}^{(k)} = \frac{\sqrt{P}}{\dfrac{\ell_1^{(k)}}{\Phi_1} + \dfrac{\ell_2^{(k)}}{\Phi_2}}, \quad k = 1, 2. \tag{97}$$

Reduction. The two flux equations eq. (97) are linear in the resistivities $1/\Phi_i$:

$$\frac{\ell_1^{(k)}}{\Phi_1} + \frac{\ell_2^{(k)}}{\Phi_2} = \frac{\sqrt{P}}{J_{\text{ss}}^{(k)}}, \quad k = 1, 2. \tag{98}$$

Since the coefficient matrix $\left(\begin{smallmatrix} \ell_1^{(1)} & \ell_2^{(1)} \\ \ell_1^{(2)} & \ell_2^{(2)} \end{smallmatrix}\right)$ is nonsingular whenever the thickness ratios differ, this 2×2 linear system determines Φ_1 and Φ_2 uniquely.

With Φ_1, Φ_2 known, the two timelag equations eq. (96) become two equations in three unknowns (D_1, D_2, K). However, $\mathrm{K} = S_2/S_1 = (\Phi_2/D_2)/(\Phi_1/D_1) = \Phi_2 D_1/(\Phi_1 D_2)$, reducing the system to two equations in two unknowns (D_1, D_2), which is generically solvable. The solubilities then follow: $S_i = \Phi_i/D_i$.

Proposition 4.4 (Solvability of Method 2)**.** *Two bilayer samples with distinct thickness ratios and known layer thicknesses generically determine all four parameters D_1, D_2, S_1, S_2 from the measured timelags and steady-state fluxes.*

Remark 4.5. With $N > 2$ samples, the system is over-determined. A least-squares approach

$$\min_{D_1, D_2, S_1, S_2 > 0} \sum_{k=1}^{N} \left[w_\tau \Big(\tau_{\text{meas}}^{(k)} - \tau_{\text{model}}^{(k)}\Big)^2 + w_J \Big(J_{\text{ss,meas}}^{(k)} - J_{\text{ss,model}}^{(k)}\Big)^2 \right] \tag{99}$$

(with suitable weights w_τ, w_J) provides uncertainty estimates and can detect systematic discrepancies indicating, e.g., interface resistance or concentration-dependent diffusivity.

4.4 Method 3: Full Transient Curve Fitting

The cumulative permeation curve $Q(t)$ contains more information than the three asymptotic parameters (τ_{lag}, J_{ss}, θ_{up}). In particular, the *shape* of the transient region encodes the eigenvalue structure of the composite diffusion problem, which depends on D_1, D_2, K, ℓ_1, and ℓ_2.

Normalised transient. Define the normalised cumulative permeation

$$\hat{Q}(t) := \frac{Q(t)}{J_{\text{ss}}\, t}, \tag{100}$$

which tends to 1 at large t and whose time evolution depends only on D_1, D_2, K (and the known thicknesses), *not* on the absolute solubilities.

Four-parameter fit. Fitting the full (unnormalised) $Q(t)$ to the analytical model (eq. (55) inverted numerically, or equivalently a numerical PDE solution) determines all four parameters simultaneously:

- The transient shape pins down D_1, D_2, and K.
- The absolute magnitude of J_{ss} then determines S_1 (and hence $S_2 = \mathrm{K}\, S_1$) via eq. (68).

Remark 4.6. This method extracts all four parameters from a *single* experiment but requires high-quality time-resolved data throughout the transient. It is also more computationally demanding and may be sensitive to noise, particularly when the two layers have similar properties. Regularisation or Bayesian inference approaches can improve robustness. Numerical validation with FESTIM is recommended to verify the extracted parameters.

4.5 Special Case: Known Solubilities

When the solubility coefficients S_1 and S_2 are known from independent measurements (literature values, separate sorption experiments, or electrochemical methods), the problem reduces significantly: only D_1 and D_2 are unknown, and a single bilayer experiment suffices.

Method A: Timelag + steady-state flux. From eq. (68), with $\Phi_i = D_i S_i$:

$$\frac{\ell_1}{D_1 S_1} + \frac{\ell_2}{D_2 S_2} = \frac{\sqrt{P}}{J_{\text{ss}}}. \tag{101}$$

This is one linear equation in $1/D_1$ and $1/D_2$. The composite timelag eq. (71) provides a second (nonlinear) equation.

Define $u_i \coloneqq 1/D_i$. From eq. (101):

$$u_2 = \frac{S_2}{\ell_2}\left(\frac{\sqrt{P}}{J_{\text{ss}}} - \frac{\ell_1}{S_1}\, u_1\right). \tag{102}$$

Substituting into eq. (71) yields a single equation in u_1, solvable numerically.

Method B: Timelag + upstream lead time. If θ_{up} is measurable, the pair $(\tau_{\text{lag}}, \theta_{\text{up}})$ provides two equations in (D_1, D_2) with $\mathrm{K} = S_2/S_1$ known. This has the advantage of relying only on *time-domain* intercepts, avoiding the need for absolute flux calibration.

4.6 Dimensionless Diagnostics and Conditioning

Rather than examining the six individual parameters (D_1, D_2, S_1, S_2, ℓ_1, ℓ_2) separately, the conditioning of the extraction is governed by two dimensionless ratios that capture the *relative* contributions of the two layers.

4.6.1 The Governing Dimensionless Groups

Definition 4.7 (Resistance ratio)**.** The *resistance ratio* is

$$\kappa \coloneqq \frac{R_1}{R_2} = \frac{\ell_1/\Phi_1}{\ell_2/\Phi_2} = \frac{\ell_1\, D_2\, S_2}{\ell_2\, D_1\, S_1}, \tag{103}$$

where $R_i = \ell_i/\Phi_i = \ell_i/(D_i S_i)$ is the permeation resistance of layer i.

κ determines **which layer controls the steady-state flux**. The fractional resistance of Layer 1 is $\eta = \kappa/(1+\kappa)$. When $\kappa \gg 1$, Layer 1 dominates (high-resistance barrier); when $\kappa \ll 1$, Layer 2 dominates.

Definition 4.8 (Diffusion speed ratio)**.** The *diffusion speed ratio* is

$$\rho := \frac{D_1\,\ell_2}{D_2\,\ell_1}. \tag{104}$$

ρ compares the diffusive transit speeds of the two layers: since the single-layer timelag scales as $\ell_i^2/(6D_i)$, $\rho \gg 1$ means Layer 1 is "fast" (short transit time relative to its thickness), while $\rho \ll 1$ means Layer 1 is the diffusion bottleneck.

Remark 4.9. The two ratios encode all six parameters. In particular, $\kappa = \mathrm{K}/\rho$ where $\mathrm{K} = S_2/S_1$ is the partition coefficient. Thus any pair among $\{\kappa, \rho, \mathrm{K}\}$ suffices to characterise the bilayer, together with a single reference scale (e.g., ℓ_1^2/D_1 for time).

4.6.2 Dimensionless Timelag

The composite timelag from eq. (70) takes a transparent form when non-dimensionalised as $\hat{\tau} = \tau_{\mathrm{lag}}\, D_1/\ell_1^2$:

$$\boxed{\hat{\tau} = \frac{\frac{1}{6} + \frac{\rho^2}{2} + \frac{1}{\kappa}\left(\frac{\rho^2}{6} + \frac{1}{2}\right)}{1 + \frac{1}{\kappa}}.} \tag{105}$$

Each term in the numerator has a direct physical interpretation:

Term	**Physical origin**
$\frac{1}{6}$	Layer 1 self-timelag ($\propto \ell_1^2/D_1$)
$\frac{\rho^2}{2}$	Cross-loading: Layer 2 filled through Layer 1
$\frac{1}{\kappa} \cdot \frac{\rho^2}{6}$	Layer 2 self-timelag ($\propto \ell_2^2/D_2$)
$\frac{1}{\kappa} \cdot \frac{1}{2}$	Cross-loading: Layer 1 filled through Layer 2

The denominator $1 + 1/\kappa$ is the total dimensionless resistance. The cross-loading terms are the signature of the bilayer: they couple information about *both* layers into the timelag, even when one layer dominates the self-timelag contribution.

4.6.3 Regime Analysis

Whether the extraction is well- or ill-conditioned depends on whether *both* τ_{lag} and J_{ss} retain sensitivity to *both* layers. Table 1 summarises the key regimes.

Table 1: Extraction conditioning as a function of the dimensionless ratios κ and ρ.

Regime	κ	ρ	τ_{lag} **sees both?**	J_{ss} **sees both?**	**Extraction**
Both layers visible	$\sim$0.1–10	$\sim$0.1–10	Yes	Yes	Well-conditioned
Thin, fast Layer 1	$\ll 1$	$\gg 1$	No	No	Poor
Thin barrier (Layer 1)	$\gg 1$	any	Yes (cross-loading)	Yes (dominates)	Well-conditioned
Identical materials	$= 1$	$= 1$, $\mathrm{K} = 1$	Degenerate	Degenerate	Impossible
One very slow layer	~ 1	$\ll 1$ or $\gg 1$	Partially	Yes	Marginal

Why the thin, fast layer case fails. When $\kappa \ll 1$ and $\rho \gg 1$, Layer 1 is thin and diffuses rapidly. Its self-timelag $\propto \ell_1^2/D_1$ is negligible, and its resistance $R_1 = \ell_1/\Phi_1$ is a small fraction of the total. Both the timelag and the flux are dominated by Layer 2; changing D_1 or S_1 barely alters either observable. Physically, Layer 1 equilibrates almost instantaneously and acts as a transparent window.

Why the fast/slow layer case is marginal. When $\rho \gg 1$ with $\kappa \sim 1$ (comparable resistances but very different diffusivities), the timelag reduces to $\hat{\tau} \approx \rho^2/2 + (1/\kappa)\rho^2/6$, which depends on $\rho^2 = (D_1/D_2)^2 \cdot (\ell_2/\ell_1)^2$. The fast layer's diffusivity D_1 cancels in the leading term after dividing by d_0:

$$\tau_{\text{lag}} \approx \frac{\ell_2^2}{6D_2} \quad \text{when } \rho \gg 1,\ \kappa \sim 1. \tag{106}$$

In other words, the timelag is blind to D_1 because Layer 1 equilibrates instantaneously regardless of whether D_1 is 10× or 1000× larger than D_2. The flux J_{ss} still depends on $\Phi_1 = D_1 S_1$ through the series resistance, providing partial but not full information.

4.6.4 The Thin Barrier Coating Case ($\kappa \gg 1$)

A particularly important practical scenario is a thin, low-permeability coating (Layer 1) on a thick, more permeable substrate (Layer 2). Here $\ell_1 \ll \ell_2$ but $\Phi_1 \ll \Phi_2$, so $\kappa \gg 1$. Despite the small thickness, the extraction is **well-conditioned**:

Flux sensitivity. Since $\kappa \gg 1$, the barrier dominates the total resistance:

$$J_{\text{ss}} \approx \frac{\sqrt{P}\,\Phi_1}{\ell_1} \qquad (\kappa \gg 1). \tag{107}$$

The flux is almost entirely determined by the coating's permeability Φ_1, providing high sensitivity to $D_1 S_1$.

Timelag sensitivity. The composite timelag in this limit is (restoring dimensions):

$$\tau_{\text{lag}} \approx \frac{\ell_1^2}{6D_1} + \frac{\mathrm{K}\,\ell_1\,\ell_2}{2D_1} + \frac{\mathrm{K}\,\ell_2^2}{6D_2}, \qquad (\kappa \gg 1). \tag{108}$$

The middle term—the *cross-loading* contribution—is the key: it represents the time required to load the thick substrate *through* the barrier bottleneck. Even though ℓ_1 is small, this term scales as $\ell_1\ell_2/D_1$, and because ℓ_2 is large and D_1 is small (barrier property), it can dominate the timelag. This ensures that D_1 remains visible in the timelag measurement despite the barrier being thin.

Physical picture. In a barrier-coated membrane, hydrogen must pass through the thin but impermeable coating to reach the thick substrate. The substrate acts as a large reservoir that must be "filled" through the coating bottleneck before steady state is reached. The filling time depends explicitly on D_1 (the coating diffusivity), keeping it experimentally accessible.

Exception. If $\kappa \gg 1$ and $\rho \gg 1$ *simultaneously*, the layer has high resistance *because of very low solubility* ($S_1 \ll S_2$) rather than low diffusivity. In this unusual case, the barrier diffuses quickly but dissolves very little hydrogen; the cross-loading term in eq. (108) becomes small because $\mathrm{K} = S_2/S_1$ and D_1 are both large, and their ratio K/D_1 may not be. This is atypical for practical barrier materials, which generally have *both* low D_i and low S_i.

4.6.5 Practical Diagnostic

Before attempting extraction, a simple check can assess feasibility:

1. **Compare timelags.** If a single-layer measurement of the substrate (Layer 2) is available, compute $\tau_2^* = (\ell_2^*)^2/(6D_2)$. If the bilayer timelag τ_{lag} differs significantly from $\tau_2^* \cdot (\ell_2/\ell_2^*)^2$ (adjusting for thickness), then Layer 1 contributes measurably and extraction is feasible.

2. **Estimate κ.** Using literature values or order-of-magnitude estimates for Φ_i, compute $\kappa = (\ell_1/\Phi_1)/(\ell_2/\Phi_2)$. Values in the range $10^{-2} < \kappa < 10^2$ (with κ not too close to 1 if $\mathrm{K} \approx 1$) indicate a well-conditioned regime.

3. **Estimate ρ.** Compute $\rho = D_1\ell_2/(D_2\ell_1)$. The case $\kappa \gg 1$ with $\rho \lesssim 1$ (true barrier: low D, low S) is the most favourable for extraction.

Proposition 4.10 (Condition for well-posed extraction, refined)**.** *The extraction of individual parameters is well-conditioned when either:*

1. *Both layers contribute comparably to the total resistance ($\kappa \sim 1$) and to the timelag ($\rho \sim 1$), with the layers being materially distinct ($\mathrm{K} \neq 1$ or $\rho \neq 1$); or*

2. *One layer dominates the resistance ($\kappa \gg 1$ or $\kappa \ll 1$)* ***and*** *the cross-loading term in the timelag remains significant—as is the case for thin barrier coatings where both D_i and S_i of the barrier are small.*

The extraction degenerates when both τ_{lag} and J_{ss} lose sensitivity to a layer simultaneously, which occurs for thin, fast, highly permeable layers ($\kappa \ll 1$, $\rho \gg 1$) or for identical materials ($\kappa = \rho = 1$, $\mathrm{K} = 1$).

4.7 Summary of Methods

Table 2: Extraction methods for individual layer transport parameters.

Method	**Required data**	**Unknowns**	**Notes**
1. Single-layer cal. + bilayer	$(\tau_i^*, J_i^*) + (\tau_{\mathrm{lag}}, J_{\mathrm{ss}})$	4	Most robust
2. Two bilayer samples	$(\tau_{\mathrm{lag}}^{(k)}, J_{\mathrm{ss}}^{(k)})$, $k = 1, 2$	4	No single-layer needed
3. Full transient fit	$Q(t)$ (full curve)	4	Single experiment
A. Known S_i: $\tau_{\mathrm{lag}} + J_{\mathrm{ss}}$	τ_{lag}, J_{ss}	2	Simplest
B. Known S_i: $\tau_{\mathrm{lag}} + \theta_{\mathrm{up}}$	τ_{lag}, θ_{up}	2	No flux cal. needed

4.8 Practical Protocol

Based on the preceding analysis, we recommend the following experimental procedure for extracting individual transport parameters from a bilayer membrane when solubilities are *not* known a priori.

1. **Sample preparation.** Fabricate (or obtain):

   - A bilayer membrane with known ℓ_1, ℓ_2 (verified by cross-sectional microscopy if possible).
   - *Either* a single-layer sample of one constituent material (Method 1) *or* a second bilayer sample with a different thickness ratio (Method 2).

2. **Single-layer calibration** (if using Method 1). Conduct a gas-driven permeation experiment on the single-layer sample:

   - Extract τ_i^* and J_{ss}^* from the $Q(t)$ asymptote.
   - Compute D_i and S_i via eqs. (93) and (94).

3. **Bilayer permeation experiment(s).**

- Apply a known upstream hydrogen pressure P.
- Record $Q(t)$ until well into steady state.
- Extract τ_{lag} (intercept) and J_{ss} (slope) from the linear asymptote.
- Optionally, record the upstream pressure decay for θ_{up} measurement.

4. **Parameter extraction.**
   - **Method 1:** Use the known D_i, S_i together with bilayer τ_{lag} and J_{ss} to solve for D_j, S_j (section 4.2).
   - **Method 2:** Use both bilayer datasets to first determine Φ_1, Φ_2 from the flux system, then D_1, D_2 from the two timelags (section 4.3).
   - **Method 3:** Fit the full transient $Q(t)$ to the analytical model (section 4.4).
5. **Validation.** Using the extracted parameters:
   - Reconstruct the full transient $Q(t)$ curve using a FESTIM simulation or numerical inversion of the Laplace-domain solution.
   - Compare the reconstructed curve with the measured data.
   - Good agreement validates both the extracted parameters and the bilayer diffusion model; systematic discrepancies may indicate interface resistance, concentration-dependent diffusivity, or trapping effects.

5 Conclusions and Future Work

5.1 Summary of Results

We have presented a self-contained derivation of the gas-driven permeation timelag for a bilayer composite membrane using Laplace-transform methods. The principal results are:

1. **Composite timelag formula.** The downstream timelag for a bilayer membrane with layer thicknesses ℓ_1, ℓ_2, diffusivities D_1, D_2, and solubilities S_1, S_2 is given by eq. (71). This reduces to the classical $L^2/(6D)$ result for a homogeneous membrane and agrees with the dimensionless form of Tran, Meldon, and Sontag [7].
2. **Direction independence of the downstream timelag.** The downstream timelag is independent of the direction of permeation (theorem 3.3). This has the important practical consequence that measuring the downstream timelag in both the forward and reverse directions does not yield two independent equations for parameter extraction.
3. **Direction dependence of the upstream lead time.** In contrast to the downstream timelag, the upstream lead time θ_{up} is direction-dependent (theorem 3.5), providing additional independent information.
4. **Direction independence of the steady-state flux.** Like the timelag, the steady-state flux through a bilayer is independent of the permeation direction, depending only on the layer permeabilities $\Phi_i = D_i S_i$ and $\sqrt{P}$.
5. **Extraction methods.** We showed that a single permeation experiment on a bilayer yields at most three independent quantities (τ_{lag}, J_{ss}, θ_{up}), which are insufficient to determine all four unknowns (D_1, D_2, S_1, S_2) without additional information. Three strategies are proposed (section 4):
   - *Method 1*: Single-layer calibration of one constituent material, combined with a bilayer experiment.

- *Method 2*: Two bilayer samples with different thickness ratios, requiring no single-layer data.
- *Method 3*: Full transient curve fitting to extract all parameters from a single experiment.

When solubilities are known a priori, the problem reduces to two unknowns and a single experiment suffices. The extraction is well-conditioned when both layers contribute comparably to the total permeation resistance (theorem 4.10).

5.2 Limitations

The analysis presented here rests on several assumptions:

- Fickian diffusion with constant, concentration-independent diffusivities in each layer.
- Ideal boundary conditions: instantaneous equilibration at the external surfaces (no surface resistance).
- Ideal interface: thermodynamic equilibrium and flux continuity at the layer interface (no interfacial resistance).
- Known layer thicknesses ℓ_1, ℓ_2 (solubilities need not be known a priori, but additional measurements are then required).

Departures from these assumptions—such as concentration-dependent diffusivity, surface kinetic barriers, interfacial resistance, or hydrogen trapping—will modify the timelag expressions and may require extended models.

5.3 Future Directions

Several extensions of the present work are of practical interest:

1. **Three or more layers.** The Laplace-transform approach generalises naturally to N-layer composites via transfer-matrix methods, yielding timelag expressions as functions of $2N$ material parameters (D_i, S_i).
2. **Cylindrical and spherical geometries.** Pipeline and pressure vessel applications involve cylindrical membranes, where the curvature modifies the steady-state concentration profile and the timelag expression.
3. **Concentration-dependent diffusivity.** In many polymers and some metals, D depends on concentration. The timelag then depends on the upstream concentration C_0, and multiple experiments at different pressures can provide additional constraints.
4. **Trapping effects.** In metals, hydrogen trapping at microstructural defects increases the effective timelag. Incorporating McNabb–Foster or Oriani trapping models into the bilayer framework is an important extension.
5. **Numerical validation.** The analytical results should be validated against finite-element simulations (e.g., using FESTIM or COMSOL) with realistic material parameters, providing confidence in both the derivations and the extraction algorithms under practical noise levels.

A Detailed Laplace Transform Calculations

This appendix provides the detailed algebraic steps omitted from the main text in the interest of readability.

A.1 Simplification of the Coefficient E

From eq. (40), $E = \mathrm{K}[A\cosh(q_1\ell_1) + B\sinh(q_1\ell_1)]$. Substituting $A = C_0/s$ and the expression for B from eq. (46):

$$\begin{aligned} E &= \mathrm{K}\left[\frac{C_0}{s}\cosh(q_1\ell_1) - \frac{C_0}{s}\,\frac{D_1\,q_1\sinh(q_1\ell_1) + R\cosh(q_1\ell_1)}{D_1\,q_1\cosh(q_1\ell_1) + R\sinh(q_1\ell_1)}\,\sinh(q_1\ell_1)\right] \\ &= \frac{\mathrm{K}\,C_0}{s}\,\frac{\cosh(q_1\ell_1)\big[D_1\,q_1\cosh(q_1\ell_1) + R\sinh(q_1\ell_1)\big] - \sinh(q_1\ell_1)\big[D_1\,q_1\sinh(q_1\ell_1) + R\cosh(q_1\ell_1)\big]}{D_1\,q_1\cosh(q_1\ell_1) + R\sinh(q_1\ell_1)}. \end{aligned} \tag{109}$$

The numerator bracket simplifies using $\cosh^2 z - \sinh^2 z = 1$:

$$\begin{aligned} &D_1\,q_1\big[\cosh^2(q_1\ell_1) - \sinh^2(q_1\ell_1)\big] + R\big[\sinh(q_1\ell_1)\cosh(q_1\ell_1) - \cosh(q_1\ell_1)\sinh(q_1\ell_1)\big] \\ &= D_1\,q_1\cdot 1 + R\cdot 0 = D_1\,q_1. \end{aligned} \tag{110}$$

Therefore,

$$E = \frac{\mathrm{K}\,C_0\,D_1\,q_1}{s\big[D_1\,q_1\cosh(q_1\ell_1) + R\sinh(q_1\ell_1)\big]}, \tag{111}$$

confirming eq. (51).

A.2 Full Small-s Expansion of the Denominator

We expand the denominator function

$$\mathcal{D}(s) = D_1\,q_1\cosh(q_1\ell_1)\sinh(q_2\ell_2) + \mathrm{K}\,D_2\,q_2\sinh(q_1\ell_1)\cosh(q_2\ell_2) \tag{112}$$

to order s^2. Recall $q_i = \sqrt{s/D_i}$ and $\alpha_i = q_i\ell_i$, so $\alpha_i^2 = s\ell_i^2/D_i$.

Term 1: $D_1\,q_1\cosh(\alpha_1)\sinh(\alpha_2)$.

$$D_1\,q_1 = \sqrt{D_1\,s}, \tag{113}$$

$$\cosh(\alpha_1) = 1 + \frac{\alpha_1^2}{2} + \frac{\alpha_1^4}{24} + O(\alpha_1^6) = 1 + \frac{s\ell_1^2}{2D_1} + \frac{s^2\ell_1^4}{24D_1^2} + O(s^3), \tag{114}$$

$$\sinh(\alpha_2) = \alpha_2 + \frac{\alpha_2^3}{6} + \frac{\alpha_2^5}{120} + O(\alpha_2^7) = \ell_2\sqrt{\frac{s}{D_2}}\left(1 + \frac{s\ell_2^2}{6D_2} + \frac{s^2\ell_2^4}{120D_2^2} + O(s^3)\right). \tag{115}$$

Product:

$$\begin{aligned} &D_1\,q_1\cosh(\alpha_1)\sinh(\alpha_2) \\ &= \sqrt{D_1}\cdot\frac{\ell_2\,s}{\sqrt{D_2}}\left(1 + \frac{s\ell_1^2}{2D_1} + \frac{s^2\ell_1^4}{24D_1^2} + \cdots\right)\left(1 + \frac{s\ell_2^2}{6D_2} + \frac{s^2\ell_2^4}{120D_2^2} + \cdots\right) \\ &= \frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}}\,s\left[1 + s\left(\frac{\ell_1^2}{2D_1} + \frac{\ell_2^2}{6D_2}\right) + s^2\left(\frac{\ell_1^4}{24D_1^2} + \frac{\ell_1^2\ell_2^2}{12D_1D_2} + \frac{\ell_2^4}{120D_2^2}\right) + O(s^3)\right]. \end{aligned} \tag{116}$$

Term 2: $\mathrm{K}\,D_2\,q_2\sinh(\alpha_1)\cosh(\alpha_2)$.

By the same procedure:

$$\begin{aligned} &\mathrm{K}\,D_2\,q_2\sinh(\alpha_1)\cosh(\alpha_2) \\ &= \frac{\mathrm{K}\ell_1\sqrt{D_2}}{\sqrt{D_1}}\,s\left[1 + s\left(\frac{\ell_1^2}{6D_1} + \frac{\ell_2^2}{2D_2}\right) + s^2\left(\frac{\ell_1^4}{120D_1^2} + \frac{\ell_1^2\ell_2^2}{12D_1D_2} + \frac{\ell_2^4}{24D_2^2}\right) + O(s^3)\right]. \end{aligned} \tag{117}$$

Sum. Combining eqs. (116) and (117) and collecting powers of s:

$$\mathcal{D}(s) = d_0\, s + d_1\, s^2 + d_2\, s^3 + O(s^4), \tag{118}$$

where

$$d_0 = \frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}} + \frac{\mathrm{K}\ell_1\sqrt{D_2}}{\sqrt{D_1}} = \frac{\ell_2 D_1 + \mathrm{K}\ell_1 D_2}{\sqrt{D_1 D_2}}, \tag{119}$$

$$\begin{aligned} d_1 &= \frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}}\left(\frac{\ell_1^2}{2D_1} + \frac{\ell_2^2}{6D_2}\right) + \frac{\mathrm{K}\ell_1\sqrt{D_2}}{\sqrt{D_1}}\left(\frac{\ell_1^2}{6D_1} + \frac{\ell_2^2}{2D_2}\right) \\ &= \frac{1}{\sqrt{D_1 D_2}}\left[\frac{\ell_1^2\ell_2}{2} + \frac{\ell_2^3 D_1}{6D_2} + \frac{\mathrm{K}\ell_1^3 D_2}{6D_1} + \frac{\mathrm{K}\ell_1\ell_2^2}{2}\right], \end{aligned} \tag{120}$$

$$\begin{aligned} d_2 &= \frac{\ell_2\sqrt{D_1}}{\sqrt{D_2}}\left(\frac{\ell_1^4}{24D_1^2} + \frac{\ell_1^2\ell_2^2}{12D_1D_2} + \frac{\ell_2^4}{120D_2^2}\right) \\ &\quad + \frac{\mathrm{K}\ell_1\sqrt{D_2}}{\sqrt{D_1}}\left(\frac{\ell_1^4}{120D_1^2} + \frac{\ell_1^2\ell_2^2}{12D_1D_2} + \frac{\ell_2^4}{24D_2^2}\right). \end{aligned} \tag{121}$$

The timelag is $\tau_{\text{lag}} = d_1/d_0$, confirming eq. (70). The coefficient d_2 would be needed to extract higher-order transient information but does not enter the timelag or steady-state flux.

A.3 Verification of the Consistency Check

We verify that $\tau_{\text{lag}} = d_1/d_0$ reduces to $L^2/(6D)$ when $D_1 = D_2 = D$, $S_1 = S_2$ (so $\mathrm{K} = 1$).

From eq. (119):

$$d_0 = \frac{\ell_2\, D + \ell_1\, D}{D} = \ell_1 + \ell_2 = L. \tag{122}$$

From eq. (120) with $\mathrm{K} = 1$:

$$\begin{aligned} d_1 &= \frac{1}{D}\left[\frac{\ell_1^2\ell_2}{2} + \frac{\ell_2^3}{6} + \frac{\ell_1^3}{6} + \frac{\ell_1\ell_2^2}{2}\right] \\ &= \frac{1}{6D}\left[3\ell_1^2\ell_2 + \ell_2^3 + \ell_1^3 + 3\ell_1\ell_2^2\right] = \frac{(\ell_1+\ell_2)^3}{6D} = \frac{L^3}{6D}. \end{aligned} \tag{123}$$

Here we used the binomial expansion: $(\ell_1 + \ell_2)^3 = \ell_1^3 + 3\ell_1^2\ell_2 + 3\ell_1\ell_2^2 + \ell_2^3$.

Therefore $\tau_{\text{lag}} = d_1/d_0 = L^3/(6DL) = L^2/(6D)$. ✓

A.4 Reverse-Direction Flux Derivation

For the reverse direction, the feed is at $x = L$ (Layer 2 upstream) and the permeate is at $x = 0$ (Layer 1 downstream).

Boundary conditions.

$$C_2(L,t) = C_0' = S_2\sqrt{P}, \tag{124}$$

$$C_1(0,t) = 0. \tag{125}$$

Interface conditions. Same as the forward case: flux continuity and thermodynamic equilibrium at $x = \ell_1$.

Laplace-domain solution. In Layer 1 ($0 < x < \ell_1$):

$$\bar{C}_1(x,s) = A' \cosh(q_1 x) + B' \sinh(q_1 x). \tag{126}$$

The downstream boundary $\bar{C}_1(0,s) = 0$ gives $A' = 0$, so $\bar{C}_1 = B' \sinh(q_1 x)$.

In Layer 2 (using $y = x - \ell_1$, $0 < y < \ell_2$):

$$\bar{C}_2(y,s) = E' \cosh(q_2 y) + F' \sinh(q_2 y). \tag{127}$$

The upstream boundary $\bar{C}_2(\ell_2, s) = C_0'/s$ gives

$$E' \cosh(q_2 \ell_2) + F' \sinh(q_2 \ell_2) = \frac{C_0'}{s}. \tag{128}$$

Interface equilibrium: $\mathrm{K}\,\bar{C}_1(\ell_1, s) = \bar{C}_2(0,s)$, i.e.,

$$\mathrm{K}\, B' \sinh(q_1 \ell_1) = E'. \tag{129}$$

Flux continuity: $D_1\, q_1\, B' \cosh(q_1 \ell_1) = D_2\, q_2\, F'$.

Downstream flux at $x = 0$.

$$\bar{J}^{\mathrm{rev}}(0,s) = -D_1 \left.\frac{\partial \bar{C}_1}{\partial x}\right|_{x=0} = -D_1\, q_1\, B'. \tag{130}$$

(The sign convention: flux in the $-x$ direction is the downstream permeation for the reverse case. We take the magnitude.)

Solving the four-equation system for B' (details parallel to the forward case), we obtain:

$$B' = \frac{C_0'\, D_2\, q_2}{s\,\mathrm{K}\big[D_2\, q_2 \cosh(q_2\ell_2)\sinh(q_1\ell_1) + (1/\mathrm{K})\, D_1\, q_1 \sinh(q_2\ell_2)\cosh(q_1\ell_1)\big]}. \tag{131}$$

Therefore,

$$\begin{aligned}\bar{J}^{\mathrm{rev}}(0,s) &= \frac{D_1\, q_1\, C_0'\, D_2\, q_2}{s\,\mathrm{K}\big[D_2\, q_2 \cosh(q_2\ell_2)\sinh(q_1\ell_1) + (1/\mathrm{K}) D_1\, q_1 \sinh(q_2\ell_2)\cosh(q_1\ell_1)\big]} \\ &= \frac{(1/\mathrm{K})\, D_1\, D_2\, q_1\, q_2\, C_0'}{s\big[D_2\, q_2 \cosh(q_2\ell_2)\sinh(q_1\ell_1) + (1/\mathrm{K})\, D_1\, q_1 \sinh(q_2\ell_2)\cosh(q_1\ell_1)\big]}.\end{aligned} \tag{132}$$

Now multiply numerator and denominator of the fraction inside the brackets by K:

$$\bar{J}^{\mathrm{rev}}(0,s) = \frac{D_1\, D_2\, q_1\, q_2\, (C_0'/\mathrm{K})}{s\big[\mathrm{K}\, D_2\, q_2 \cosh(q_2\ell_2)\sinh(q_1\ell_1) + D_1\, q_1 \sinh(q_2\ell_2)\cosh(q_1\ell_1)\big]}. \tag{133}$$

Since $C_0'/\mathrm{K} = S_2\sqrt{P}/(S_2/S_1) = S_1\sqrt{P} = C_0$, and the denominator bracket equals

$$D_1\, q_1 \cosh(q_1\ell_1)\sinh(q_2\ell_2) + \mathrm{K}\, D_2\, q_2 \sinh(q_1\ell_1)\cosh(q_2\ell_2) = \mathcal{D}(s)/s, \tag{134}$$

(which is exactly the forward-direction denominator), we conclude that

$$\bar{J}^{\mathrm{rev}}(0,s) = \frac{\mathrm{K}\, D_1\, D_2\, q_1\, q_2\, C_0}{s\,\mathcal{D}(s)/s} \cdot \frac{1}{\mathrm{K}} = \bar{J}^{\mathrm{fwd}}(L,s). \tag{135}$$

That is, the Laplace-domain flux (and hence the cumulative permeation and timelag) is identical in both directions. This completes the detailed verification of theorem 3.3.

A.5 Upstream Lead Time: Detailed Expansion

We provide additional details for the upstream lead-time derivation of section 3.7. The upstream cumulative permeation is

$$\bar{Q}_{\mathrm{up}}(s) = \frac{\bar{J}_{\mathrm{up}}(s)}{s} = \frac{D_1\, q_1\, C_0}{s^2}\,\frac{\mathcal{N}(s)}{\mathcal{D}(s)}, \tag{136}$$

where

$$\mathcal{N}(s) = D_1\, q_1 \sinh(q_1\ell_1) + R\cosh(q_1\ell_1) \tag{137}$$

is the numerator fraction from eq. (82) (note: this $\mathcal{N}$ differs from the downstream numerator, which was simply $D_1\, q_1$).

Expanding $\mathcal{N}(s)$. Recall $R = \mathrm{K}\, D_2\, q_2 \cosh(q_2\ell_2)/\sinh(q_2\ell_2)$. For small s:

$$\begin{aligned}\frac{\cosh(q_2\ell_2)}{\sinh(q_2\ell_2)} &= \frac{1+\frac{s\ell_2^2}{2D_2}+\cdots}{\frac{\ell_2\sqrt{s}}{\sqrt{D_2}}\left(1+\frac{s\ell_2^2}{6D_2}+\cdots\right)} = \frac{\sqrt{D_2}}{\ell_2\sqrt{s}}\,\frac{1+\frac{s\ell_2^2}{2D_2}+\cdots}{1+\frac{s\ell_2^2}{6D_2}+\cdots}\\ &= \frac{\sqrt{D_2}}{\ell_2\sqrt{s}}\left(1+\frac{s\ell_2^2}{3D_2}+O(s^2)\right).\end{aligned} \tag{138}$$

Therefore,

$$R = \mathrm{K}\, D_2\sqrt{\frac{s}{D_2}}\,\frac{\sqrt{D_2}}{\ell_2\sqrt{s}}\left(1+\frac{s\ell_2^2}{3D_2}+O(s^2)\right) = \frac{\mathrm{K}\, D_2}{\ell_2}\left(1+\frac{s\ell_2^2}{3D_2}+O(s^2)\right). \tag{139}$$

Now assemble $\mathcal{N}(s)$:

$$\begin{aligned}\mathcal{N}(s) &= D_1\, q_1\sinh(q_1\ell_1) + R\cosh(q_1\ell_1)\\ &= \frac{s\ell_1^2}{\ell_1}\left(1+\frac{s\ell_1^2}{6D_1}+\cdots\right) + \frac{\mathrm{K}D_2}{\ell_2}\left(1+\frac{s\ell_2^2}{3D_2}+\cdots\right)\left(1+\frac{s\ell_1^2}{2D_1}+\cdots\right)\\ &= s\ell_1\left(1+\frac{s\ell_1^2}{6D_1}+\cdots\right) + \frac{\mathrm{K}D_2}{\ell_2}\left(1+s\left[\frac{\ell_2^2}{3D_2}+\frac{\ell_1^2}{2D_1}\right]+\cdots\right)\\ &= \frac{\mathrm{K}D_2}{\ell_2} + s\left[\ell_1+\frac{\mathrm{K}D_2}{\ell_2}\left(\frac{\ell_2^2}{3D_2}+\frac{\ell_1^2}{2D_1}\right)\right]+O(s^2)\\ &= \frac{\mathrm{K}D_2}{\ell_2} + s\left[\ell_1+\frac{\mathrm{K}\ell_2}{3}+\frac{\mathrm{K}D_2\ell_1^2}{2D_1\ell_2}\right]+O(s^2).\end{aligned} \tag{140}$$

Define

$$n_0 := \frac{\mathrm{K}D_2}{\ell_2}, \tag{141}$$

$$n_1 := \ell_1 + \frac{\mathrm{K}\ell_2}{3} + \frac{\mathrm{K}D_2\ell_1^2}{2D_1\ell_2}. \tag{142}$$

Combining with the prefactor. The full upstream cumulative is

$$\begin{aligned}\bar{Q}_{\mathrm{up}}(s) &= \frac{\sqrt{D_1\, s}\, C_0}{s^2}\,\frac{n_0+n_1\, s+O(s^2)}{d_0\, s+d_1\, s^2+O(s^3)}\\ &= \frac{C_0\sqrt{D_1}}{s^{3/2}}\,\frac{n_0+n_1\, s+O(s^2)}{d_0\, s+d_1\, s^2+O(s^3)}.\end{aligned} \tag{143}$$

Wait—let us be more careful. The prefactor in $\bar{J}_{\mathrm{up}}$ from eq. (82) is $D_1\, q_1\, C_0/s = C_0\sqrt{D_1\, s}/s = C_0\sqrt{D_1}/\sqrt{s}$. Then

$$\bar{Q}_{\mathrm{up}}(s) = \frac{C_0\sqrt{D_1}}{s\sqrt{s}}\,\frac{\mathcal{N}(s)}{\mathcal{D}(s)}. \tag{144}$$

With $\mathcal{N}(s) = n_0 + n_1\, s + O(s^2)$ and $\mathcal{D}(s) = d_0\, s + d_1\, s^2 + O(s^3)$:

$$\begin{aligned}
\bar{Q}_{\mathrm{up}}(s) &= \frac{C_0\sqrt{D_1}}{s^{3/2}}\,\frac{n_0 + n_1\, s + \cdots}{d_0\, s + d_1\, s^2 + \cdots} = \frac{C_0\sqrt{D_1}}{s^{3/2}}\,\frac{n_0 + n_1\, s + \cdots}{d_0\, s\big(1 + \frac{d_1}{d_0} s + \cdots\big)} \\
&= \frac{C_0\sqrt{D_1}}{d_0\, s^{5/2}}\big(n_0 + n_1\, s + \cdots\big)\big(1 - \tfrac{d_1}{d_0} s + \cdots\big) \\
&= \frac{C_0\sqrt{D_1}}{d_0}\left[\frac{n_0}{s^{5/2}} + \frac{n_1 - n_0\, d_1/d_0}{s^{3/2}} + O(s^{-1/2})\right]. \qquad (145)
\end{aligned}$$

The inverse Laplace transforms of fractional powers of s give $\mathcal{L}^{-1}\{s^{-5/2}\} = \frac{4t^{3/2}}{3\sqrt{\pi}}$ and $\mathcal{L}^{-1}\{s^{-3/2}\} = \frac{2t^{1/2}}{\sqrt{\pi}}$, which describe the transient approach.

However, the asymptotic (large-t) behaviour with *linear* growth comes from the poles at $s = 0$ in the full (non-expanded) expression. The correct approach is to recognise that the upstream cumulative must also approach a linear asymptote $J_{\mathrm{ss}}(t + \theta_{\mathrm{up}})$. This means we should examine the Laurent expansion more carefully.

In fact, the correct expansion uses the relation

$$\bar{Q}_{\mathrm{up}}(s) - \bar{Q}_{\mathrm{down}}(s) = \frac{\bar{M}(s)}{s}, \qquad (146)$$

where $M(t) = \int_0^L C(x,t)\,\mathrm{d}x$ is the total amount of diffusant stored in the membrane. In steady state, $M_{\mathrm{ss}} = \int_0^L C_{\mathrm{ss}}(x)\,\mathrm{d}x$ is a constant, so $\bar{M}_{\mathrm{ss}}(s) = M_{\mathrm{ss}}/s$ and

$$\bar{Q}_{\mathrm{up}}(s) = \bar{Q}_{\mathrm{down}}(s) + \frac{M_{\mathrm{ss}}}{s^2} + (\text{transient terms}). \qquad (147)$$

From the downstream expansion eq. (65):

$$\bar{Q}_{\mathrm{down}}(s) = J_{\mathrm{ss}}\left[\frac{1}{s^2} - \frac{\tau_{\mathrm{lag}}}{s} + O(1)\right]. \qquad (148)$$

Therefore, for large t:

$$Q_{\mathrm{up}}(t) \to J_{\mathrm{ss}}\, t - J_{\mathrm{ss}}\, \tau_{\mathrm{lag}} + M_{\mathrm{ss}} = J_{\mathrm{ss}}\left(t - \tau_{\mathrm{lag}} + \frac{M_{\mathrm{ss}}}{J_{\mathrm{ss}}}\right) = J_{\mathrm{ss}}(t + \theta_{\mathrm{up}}), \qquad (149)$$

with

$$\theta_{\mathrm{up}} = \frac{M_{\mathrm{ss}}}{J_{\mathrm{ss}}} - \tau_{\mathrm{lag}}. \qquad (150)$$

Computing M_{ss}. The steady-state concentration profiles are linear within each layer:

$$C_1^{\mathrm{ss}}(x) = C_0 - \frac{J_{\mathrm{ss}}}{D_1 S_1}\, S_1\, x = C_0\left(1 - \frac{x\, r_1}{\ell_1\, R_{\mathrm{tot}}}\right), \qquad 0 < x < \ell_1, \qquad (151)$$

$$C_2^{\mathrm{ss}}(y) = \mathrm{K}\, C_0\left(\frac{r_1}{R_{\mathrm{tot}}} - \frac{r_1}{R_{\mathrm{tot}}} - \frac{y}{R_{\mathrm{tot}}}\,\frac{1}{D_2 S_2/\ell_2}\right). \qquad (152)$$

More simply, at the interface $C_1^{\mathrm{ss}}(\ell_1) = C_0(1 - r_1/R_{\mathrm{tot}}) = C_0\, r_2/R_{\mathrm{tot}}$, and $C_2^{\mathrm{ss}}(0) = \mathrm{K}\, C_1^{\mathrm{ss}}(\ell_1) = \mathrm{K}\, C_0\, r_2/R_{\mathrm{tot}}$, with $C_2^{\mathrm{ss}}(\ell_2) = 0$. Each profile is linear, so

$$\begin{aligned}
M_{\mathrm{ss}} &= \int_0^{\ell_1} C_1^{\mathrm{ss}}\,\mathrm{d}x + \int_0^{\ell_2} C_2^{\mathrm{ss}}\,\mathrm{d}y \\
&= \frac{\ell_1}{2}\big(C_0 + C_0\, r_2/R_{\mathrm{tot}}\big) + \frac{\ell_2}{2}\,\mathrm{K}\, C_0\, r_2/R_{\mathrm{tot}} \\
&= \frac{C_0}{2}\left[\ell_1\left(1 + \frac{r_2}{R_{\mathrm{tot}}}\right) + \frac{\mathrm{K}\,\ell_2\, r_2}{R_{\mathrm{tot}}}\right] \\
&= \frac{C_0}{2}\left[\ell_1 + \frac{r_2}{R_{\mathrm{tot}}}\big(\ell_1 + \mathrm{K}\,\ell_2\big)\right]. \qquad (153)
\end{aligned}$$

The upstream lead time then follows from eq. (150):

$$\theta_{\mathrm{up}} = \frac{M_{\mathrm{ss}}}{J_{\mathrm{ss}}} - \tau_{\mathrm{lag}}, \tag{154}$$

where $J_{\mathrm{ss}} = C_0/R_{\mathrm{tot}}$, so $M_{\mathrm{ss}}/J_{\mathrm{ss}} = R_{\mathrm{tot}} \cdot [\text{expression in eq. (153)}]/C_0 \cdot C_0 = R_{\mathrm{tot}} \cdot [\cdots]$.

This gives an explicit, fully dimensional expression for θ_{up} in terms of the layer parameters, completing the derivation.

References

[1] R. Ash, R. M. Barrer, and J. H. Petropoulos. "Diffusion in Heterogeneous Media: Properties of a Laminated Slab". In: *British Journal of Applied Physics* 14.12 (1963), pp. 854–862. DOI: 10.1088/0508-3443/14/12/307.

[2] R. M. Barrer. *Diffusion in and Through Solids*. Cambridge: Cambridge University Press, 1951.

[3] J. Crank. *The Mathematics of Diffusion*. 2nd ed. Oxford: Oxford University Press, 1975.

[4] H. A. Daynes. "The Process of Diffusion through a Rubber Membrane". In: *Proceedings of the Royal Society of London. Series A* 97.685 (1920), pp. 286–307. DOI: 10.1098/rspa.1920.0034.

[5] H. L. Frisch. "The Time Lag in Diffusion". In: *The Journal of Physical Chemistry* 61.1 (1957), pp. 93–95. DOI: 10.1021/j150547a025.

[6] J. H. Petropoulos. "Quantitative Analysis of Gaseous Diffusion in Glassy Polymers". In: *Journal of Polymer Science: Polymer Physics Edition* 11.8 (1973), pp. 1511–1522. DOI: 10.1002/pol.1973.180110806.

[7] Hiep D. Tran, Jerry H. Meldon, and Eduardo D. Sontag. "Transient Diffusion in Bi-Layer Composites With Mass Transfer Resistance". In: *Journal of Membrane Science* 625 (2021), p. 119114. DOI: 10.1016/j.memsci.2021.119114.